

What if apps logged into you, instead of you logging into apps?

EMFcamp - June 2022



Electromagnetic Field

A brief introduction



@cpswan

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Blogger blog.thestateofme.com

Links to socials etc. at chris.swanz.net



Agenda

- How things are
- How things can be
- The IoT angle
- Inside the pipes
- You probably have questions...?
- Getting involved



How things are

Internet Architecture so far



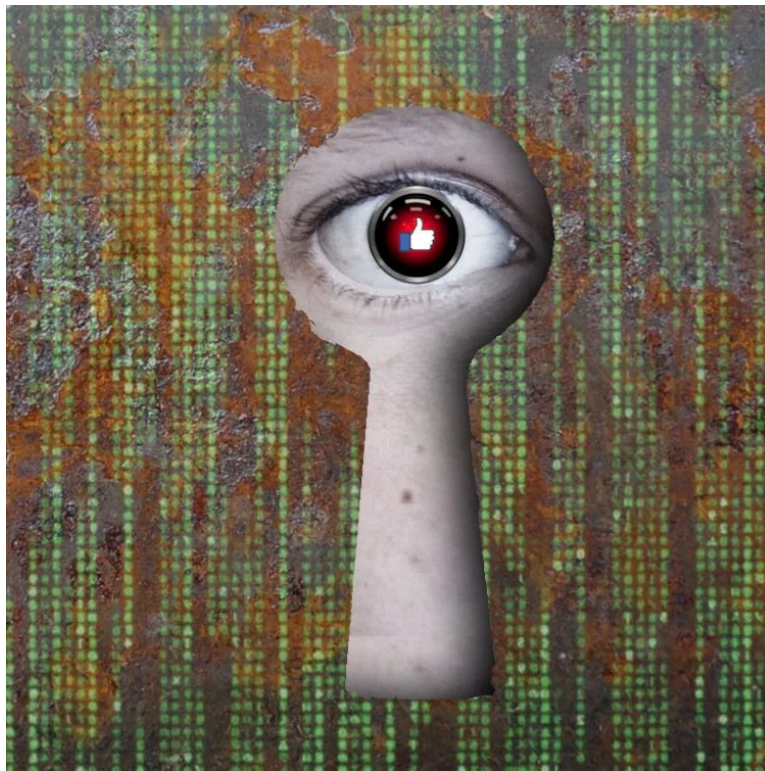


Image: [Cryteria](#); [John Lodder](#); [Denis Defreyne](#), CC BY, modified by [Cory Doctorow](#)

/the social dilemma_

<https://www.thesocialdilemma.com/>

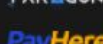
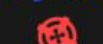
'--have i been pwned?

Check if your email or phone is in a data breach

Largest breaches

	772,904,991 Collection #1 accounts
	763,117,241 Verifications.io accounts
	711,477,622 Onliner Spambot accounts
	622,161,052 Data Enrichment Exposure From PDL Customer accounts
	593,427,119 Exploit.In accounts
	509,458,528 Facebook accounts
	457,962,538 Anti Public Combo List accounts
	393,430,309 River City Media Spam List accounts
	359,420,698 MySpace accounts
	268,765,495 Wattpad accounts

Recently added breaches

	90,655 SirHurt accounts
	112,251 Fanpass accounts
	22,424,472 Read Novel accounts
	174,168 BlackBerry Fans accounts
	348,302 OGUsers (2021 breach) accounts
	188,089 Paragon Cheats accounts
	1,580,249 PayHere accounts
	305,470 Aimware accounts
	63,451 Devil-Torrents.pl accounts
	4,101,101 Avvo accounts

<https://haveibeenpwned.com/>

How things can be

Flipping the model

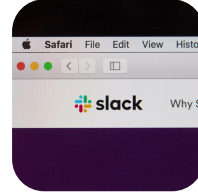
Your shopping apps



Your banking apps



Your work apps



Your social apps



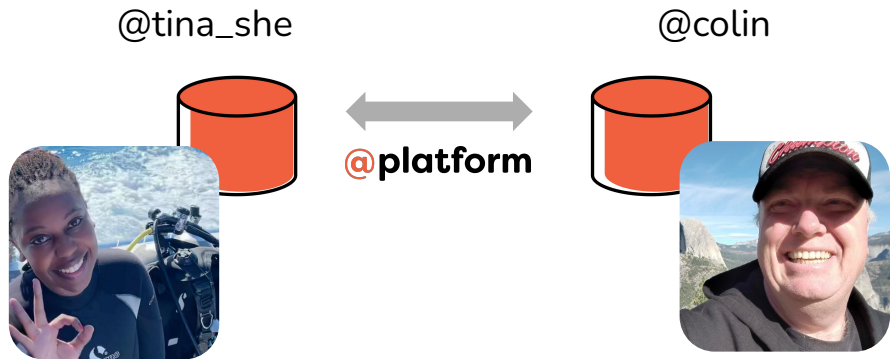
You



Your travel apps

The @platform – a change in architecture

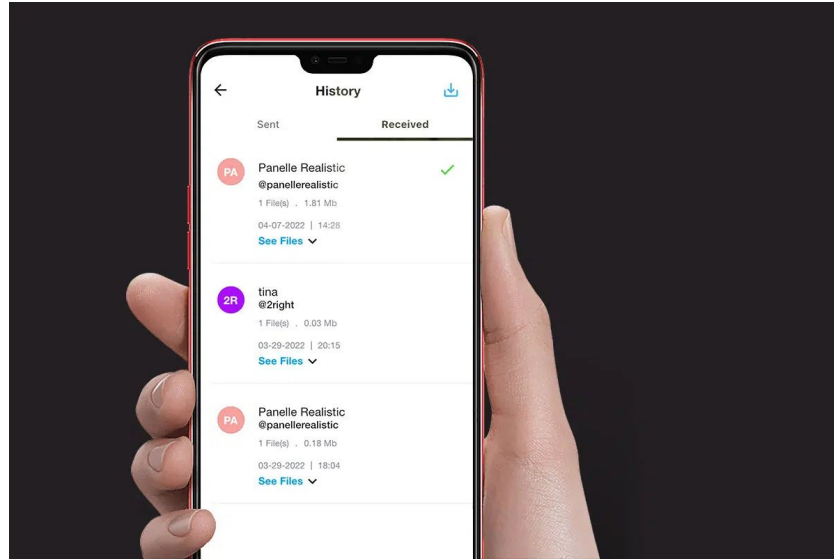
A simple open source platform and protocol – no blockchain necessary!



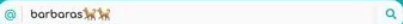
- Everyone/everything has a unique @sign
- All apps are peer-to-peer
- Microservers can be hosted anywhere
- Everything is open source
- Everything is end-to-end encrypted

Only the namespace is centralized – much like DNS

@mosphere Pro – Encrypted File Transfer



@wavi - Weblet Maker



OWNER APPROVED PUBLIC CONTENT



@barbaras 🐕🐕

NAME
Gemini and Chewy

EMAIL
GeminianandChewy@gmail.com

FOLLOW

SHARE EMBED

CURRENT LOCATION
The Dog House 🏠

SOCIAL

Instagram YouTube

DETAILS SOCIAL STREAMS FOLLOWERS (6) FOLLOWING (11)

ABOUT

Chewy and I are a couple of rescue pit bulls - well at least Chewy is a rescue, I prefer to think of myself as granting my person permission to be in my presence. I know that is very cat-like, but let's face it, some of us dogs are just better than others. Speaking of which, both of us enjoy agility competitions. I am very good and incredibly graceful. My brother, not so much.

1. HERE WE ARE ON OUR BEST BEHAVIOR:



Watch on YouTube

2

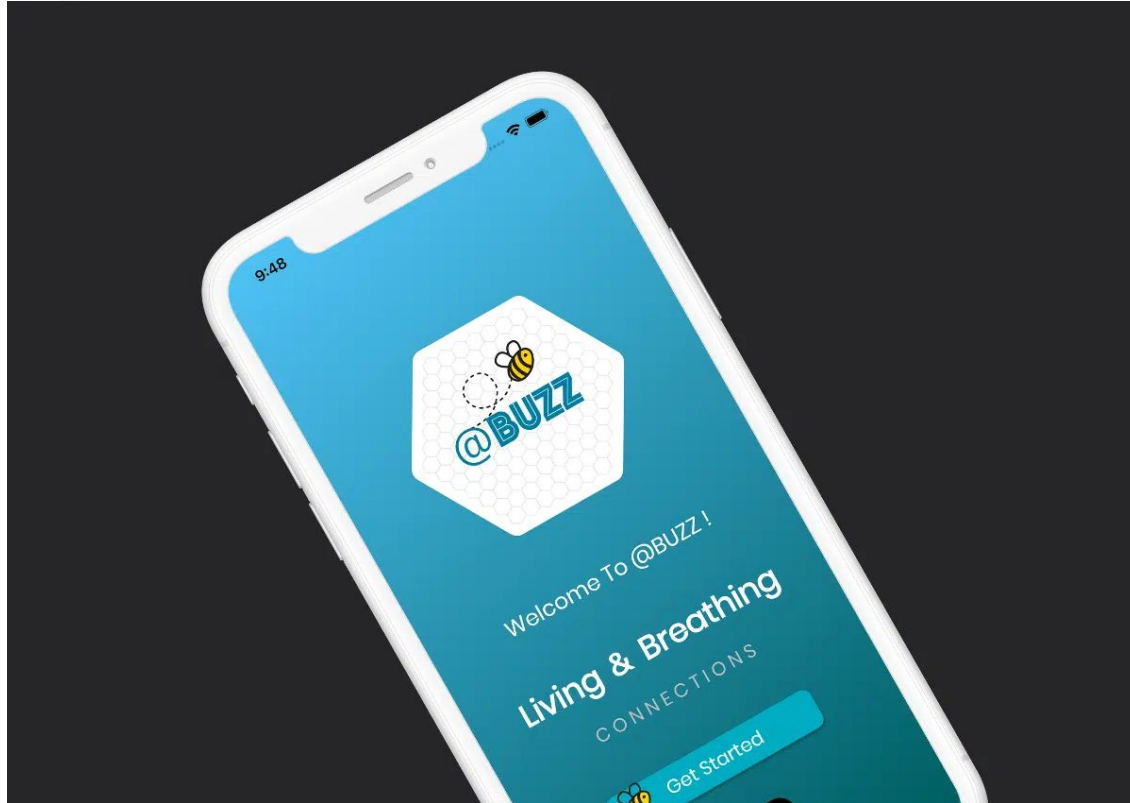
We were supposed to share our gamer tags, but Chewy and I don't play video games. That whole keyboard thing is a nonstarter with us. Our sport is called agility and that is what we do in our spare time. Here is me practicing:

3





@buzz – Real-Time Connections

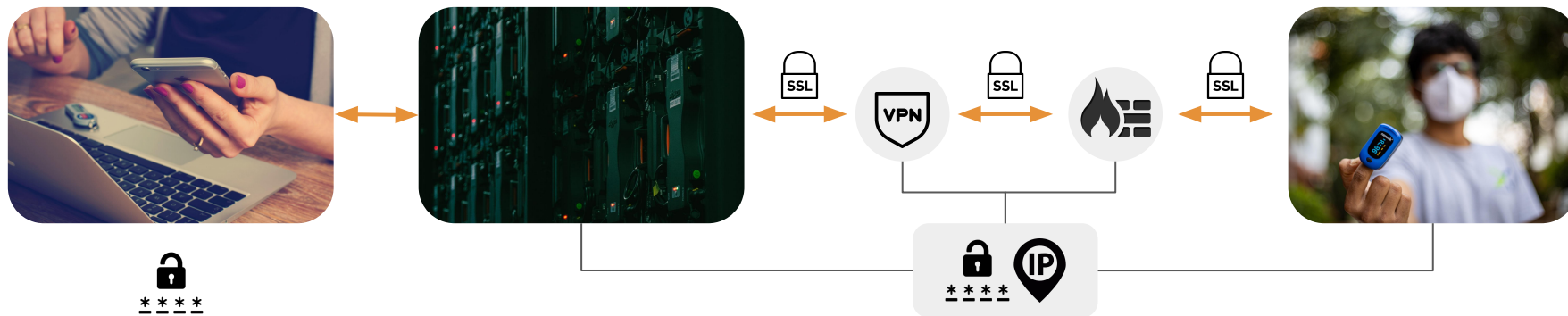


The IoT angle

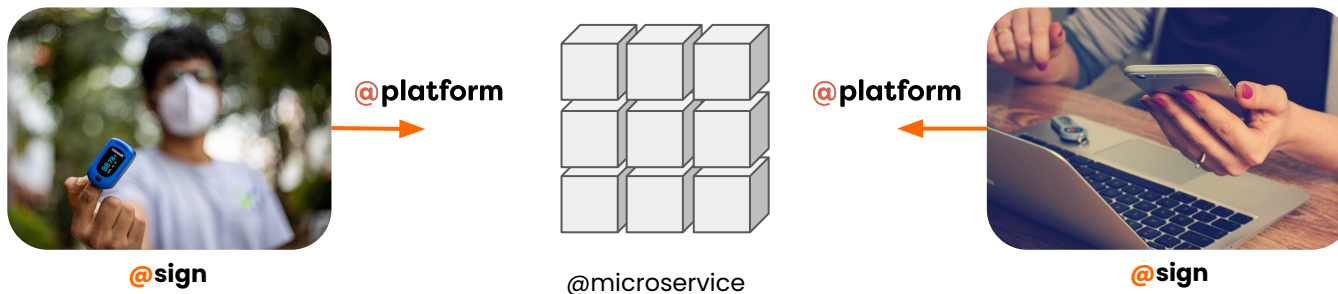
IoT flipped too



Today's IoT Architecture

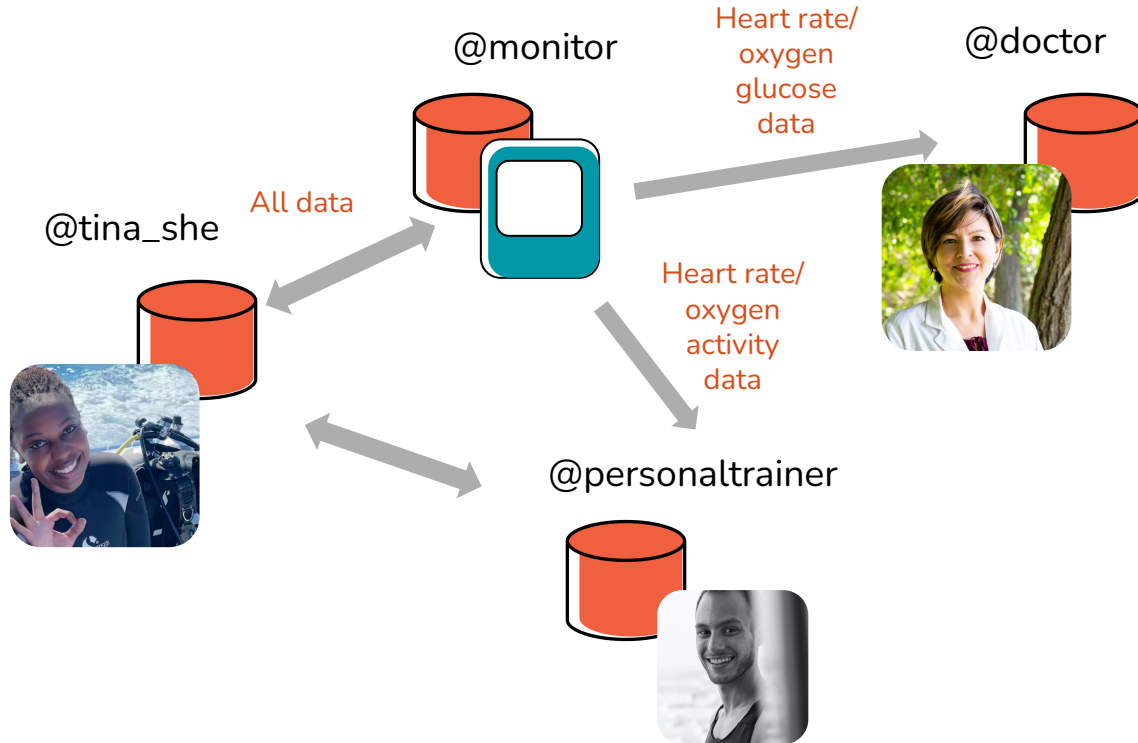


IoT flipped with the @platform



- No passwords
- No VPN/firewall
- Outbound only
- Only one open port
- No administration
- Device generates its own keys
- Device to application encryption

The @platform in action



A complete development platform that

- Protects devices
- Protects data
- Routes data

Data has both routing and attributes

- Time to live
- Time to borne/die
- Revoke

Ssh! No ports

```
cconstab@iot: ~  
cconstab@cally:~/sshnp$ $(./sshnp --from @lostblue64 \  
--to @blueamateurbinding \  
--host cally.lan \  
-d iot_device)  
Last login: Mon May  2 22:57:16 2022 from 127.0.0.1  
cconstab@iot:~$ ip a | grep 192  
    inet 192.168.1.64/24 brd 192.168.1.255 scope global dynamic ens18  
cconstab@iot:~$ logout  
cconstab@cally:~/sshnp$ nmap -p 1-65535 192.168.1.64  
Starting Nmap 7.80 ( https://nmap.org ) at 2022-05-02 23:01 UTC  
cconstab@cally:~/sshnp$  
cconstab@cally:~/sshnp$  
All 65535 scanned ports on secondary.lan (192.168.1.64) are closed  
  
Nmap done: 1 IP address (1 host up) scanned in 6.43 seconds  
cconstab@cally:~/sshnp$
```

Inside the pipes

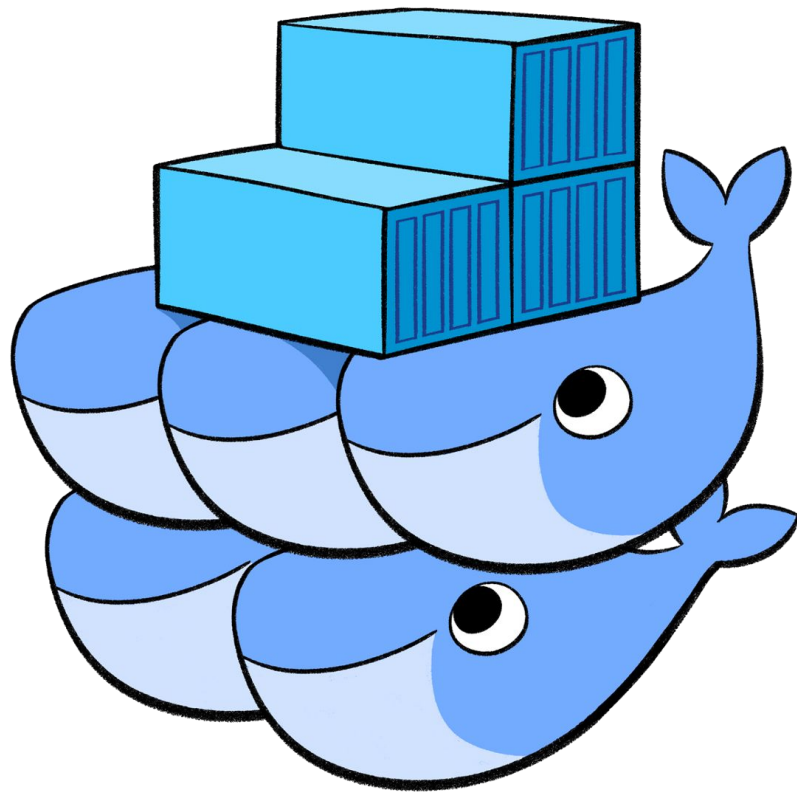
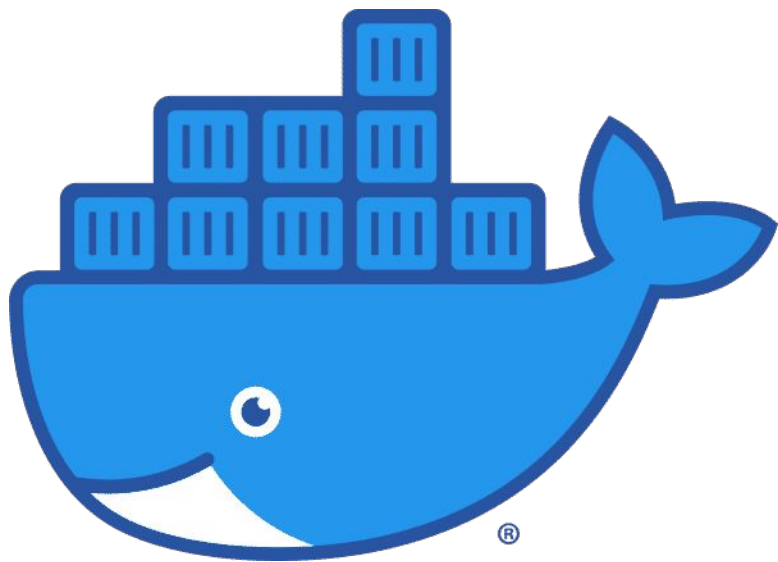
Protocol before Platform





Dart







Flutter

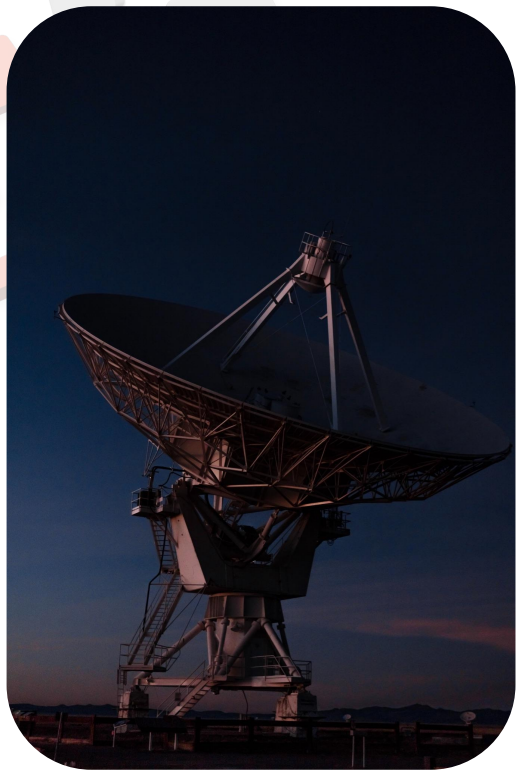




Java™



You probably have questions...?



How do we make money?

Now

- Sell custom @signs
- Host microservers (secondaries)

Future

- Offers not ads - monetizing the base with true permission-based sales
- Advanced tools/tech



A decorative graphic in the top-left corner consisting of several 3D-style squares in light red and light gray, appearing to float or fall from the top-left corner.

Who else is doing this?

- Solid (the Tim Berners-Lee thing)
 - Personal Online Data Store (PODS)
- Mydex
 - Personal Data Store
- Lots of Blockchain based attempts



Getting involved

A global, inclusive dev community



atsign.dev



No Infrastructure needed

The @platform does all the infrastructure and backend work for you.



Privacy By Design

Zero data worries for you and your consumers.



Rapid Onboarding

Cut your development time in half and launch apps faster.

pub.dev



Sign in

Search atsign.org packages

atsign.org

The @protocol is a new internet protocol that is designed to allow people to own their data and control what information they share with whom.

Dart and flutter modules for application developers that want to provide best in class privacy features can be found here. In addition, the protocol specification document and reference implementation source code can be found at atsign.dev and on GitHub at <https://github.com/atsign-foundation>.



Contributions welcome



The @ Foundation

Now for some internet optimism

📍 San Jose, CA

🔗 <https://atsign.dev>

🐦 @atsigncompany

✉️ developers@atsign.com

Verified

🏠 Overview

📁 Repositories 48

📁 Projects 4

📦 Packages

👥 Teams 1

👤 People 54

⚙️ Settings

README.md

The @ Foundation

@platform

Welcome to The @ Foundation Github Organization, home to the open source repos for [The @ Platform](#).

Time to wrap up

Review

- Things are not great - many security & privacy issues
- Things can be better, by flipping the existing model
- For things as well as for people
- Tech stack is simple and developer friendly
 - We'll go where demand takes us
- You probably have questions...?
 - Hopefully I've anticipated with some answers
- We're here because you're the people we want building stuff with us





Questions?

Contact

Chris Swan

chris@atsign.com

@cpswan



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Be yourself online.

The @ Company